

MTH302

Midterm important questions 2024

VU360

Here are some important equations questions from MTH302 Virtual University Midterm 2024:

Lecture 1 – 5:

1. Find the derivative of the function $f(x) = 3x^2 + 2x - 5$.
2. Solve the equation: $\int (2x + 1)dx = 0$
3. Find the area under the curve $y = 2x^2 + 3x - 4$ between $x = 0$ and $x = 2$.
4. Solve the differential equation: $\frac{dy}{dx} = 2x - 3y$
5. Find the volume of the solid formed by rotating the region bounded by $y = 2x^2 + 3x - 4$, $x = 0$, and $x = 2$ about the x -axis.

Lecture 6 – 10:

6. Find the Laplace transform of the function $f(t) = 2t^2 + 3t - 4$.
7. Solve the system of linear equations:
 - $2x + 3y = 5$
 - $x - 2y = -3$
8. Find the Fourier series of the function $f(x) = 2x^2 + 3x - 4$.
9. Solve the differential equation: $\frac{d^2y}{dx^2} + 4y = 0$
10. Find the area of the region bounded by $y = 2x^2 + 3x - 4$, $x = 0$, and $x = 2$.

Lecture 11 – 15:

11. Find the derivative of the function $f(x) = \frac{2x+1}{x-1}$.
12. Solve the equation: $\int \frac{\sin x}{x} dx = 0$
13. Find the volume of the solid formed by rotating the region bounded by $y = 2x^2 + 3x - 4$, $x = 0$, and $x = 2$ about the y -axis.
14. Solve the differential equation: $\frac{dy}{dx} = \frac{2x-3y}{x+y}$
15. Find the Fourier transform of the function $f(x) = 2x^2 + 3x - 4$.

Lecture 16 – 20:

16. Find the Laplace transform of the function $f(t) = t^2 - 4t + 3$.

17. Solve the system of linear equations:

- $3x + 2y = 7$
- $2x - 3y = -5$

18. Find the Fourier series of the function $f(x) = x^2 - 4x + 3$.

19. Solve the differential equation: $\frac{d^2y}{dx^2} - 4y = 0$

20. Find the area of the region bounded by $y = x^2 - 4x + 3$, $x = 0$, and $x = 2$.

Lecture 21 – 22:

21. Find the derivative of the function $f(x) = \frac{3x^2 + 2x - 1}{x - 1}$.

22. Solve the equation: $\int \frac{2x+1}{x-1} dx = 0$